

Homework 4: Streamflow Prediction with Sequence Models

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1 Problem 1: Data Access and Exploratory Analysis

The code for this assignment is available in the GitHub repository: https://github.com/nkalauni/hwrs640_rnn_hw.

1.1 Dataset Overview

The miniCAMELS dataset was accessed via the `MiniCamels` Python package. Table 1 summarizes the key dataset properties.

Table 1: miniCAMELS dataset summary.

Property	Value
Number of basins	50
Period	WY1981–WY2010 (Oct 1980 – Sep 2010)
Dynamic input variables	<code>prcp</code> , <code>tmax</code> , <code>tmin</code> , <code>srad</code> , <code>vp</code> (5 vars)
Target variable	<code>qobs</code> (mm/day)
Static attributes	16 (aridity, LAI, baseflow index, forest fraction, etc.)
Total input features	21 (5 dynamic + 16 static, per timestep)

1.2 Train/Validation/Test Split

A strictly temporal split was applied uniformly across all 50 basins:

- **Train:** WY1981–WY2003 (Oct 1, 1980 – Sep 30, 2003; 23 years)
- **Validation:** WY2004–WY2006 (Oct 1, 2003 – Sep 30, 2006; 3 years)
- **Test:** WY2007–WY2010 (Oct 1, 2006 – Sep 30, 2010; 4 years)

A temporal hold-out is the standard approach in catchment hydrology because it evaluates true out-of-sample temporal generalization, matching the operational use case of predicting future streamflow from future forcings. Holding out a contiguous future block prevents any lookahead leakage: no future observation can influence a training-set statistic or normalization parameter. Splitting by basin is also a valid approach to recreate a Prediction in Ungauged Basins (PUB) scenario. However, with only 50 basins, a spatial split would reduce the training data too much so we chose a temporal split.

1.3 Supervised Learning Sample Construction

Each sample consists of a 365-day sliding window of meteorological forcings paired with a single streamflow prediction target:

- **Input:** $\mathbf{X} \in \mathbb{R}^{365 \times 21}$ — 365 days of 5 Daymet forcings, with the 16 static basin attributes tiled along the time dimension so the model sees them at every timestep.
- **Target:** $\mathbf{qobs}$ at the final timestep of the window (one-step-ahead prediction of streamflow for day t given forcings and antecedent conditions from days $t - 364$ through t).

A 365-day window was chosen to capture a full annual hydrological cycle, including seasonal snowmelt signals and inter-annual variability in antecedent soil moisture, which are important for accurate streamflow prediction.

1.4 Exploratory Data Analysis

Figure 1 shows the distribution of observed daily streamflow across all basins and all three splits. The distribution is strongly right-skewed: the vast majority of days have low streamflow, but extreme flood events can exceed 10–100 $\times$ the median. This motivates z-score normalization of $\mathbf{qobs}$ before training rather than raw values.

Figure 2 shows daily streamflow time series for five example basins spanning the full 30-year record. The basins exhibit markedly different hydrological regimes: some show sharp spring snowmelt peaks, others show flashy summer convective responses, and still others have more subdued perennial baseflow. This diversity tests the model’s ability to generalize across climatic gradients.

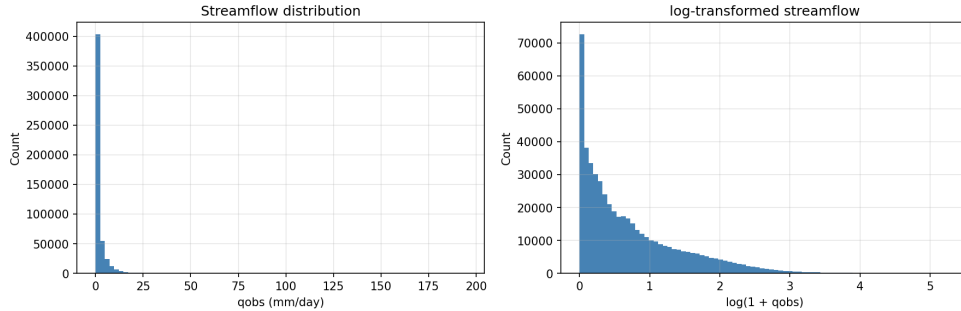


Figure 1: Distribution of observed daily streamflow ($\mathbf{qobs}$, mm/day) pooled across all 50 basins. The strong right skew motivates z-score normalization prior to model training.

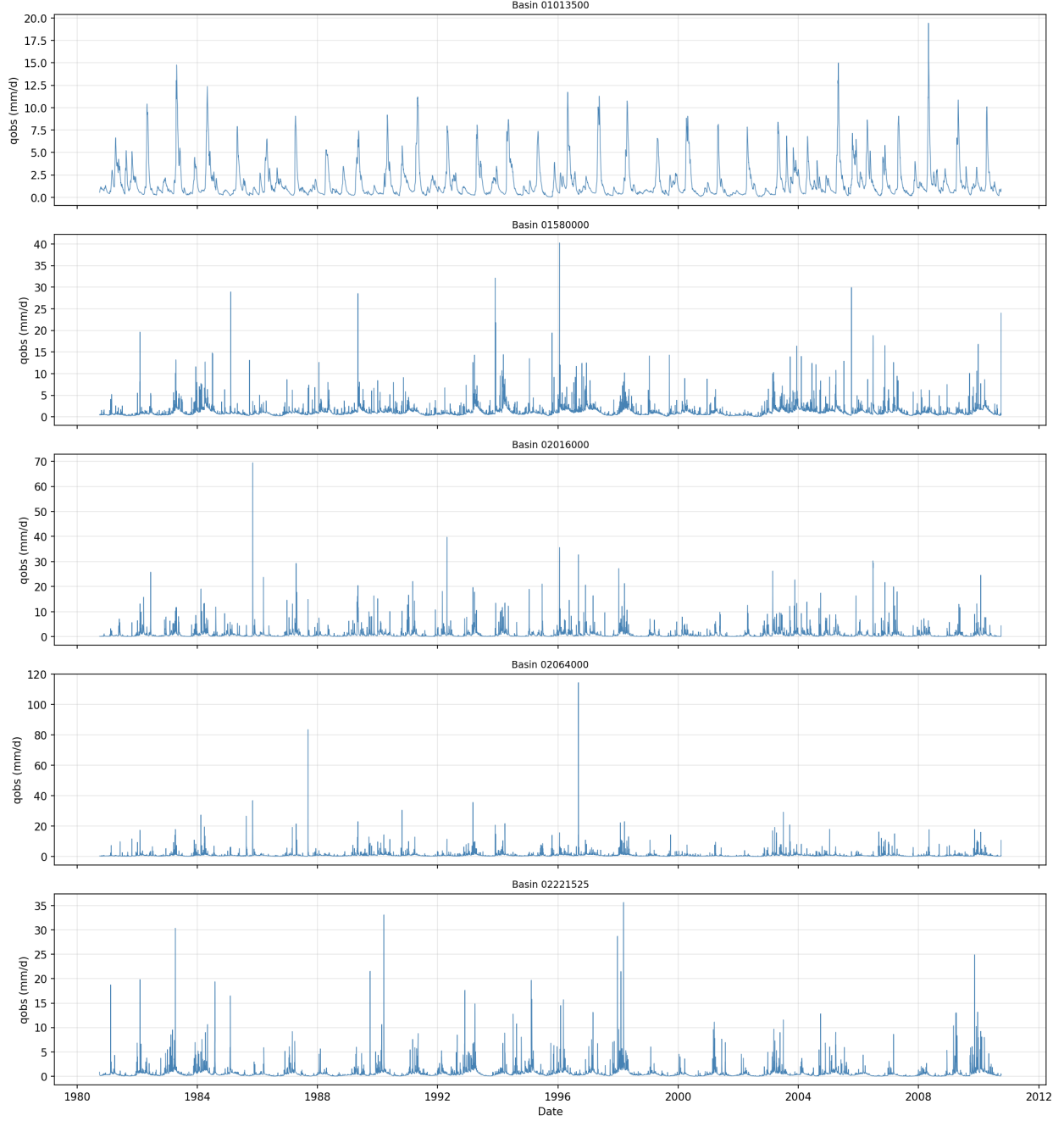


Figure 2: Daily streamflow time series for five example miniCAMELS basins (WY1981–WY2010). Vertical dashed lines mark the train/val and val/test boundaries. Basins display diverse regimes ranging from nival to flashy pluvial.

1.5 Preprocessing

Missing values: Rows with missing `qobs` were dropped before window construction. Dynamic forcings had no missing values in the dataset.

Normalization: All dynamic features, `qobs`, and static attributes were z-score normalized (μ , σ computed on the training split only) and stored as pickled `Normalizer` objects for reproducible

evaluation. Predictions are inverse-transformed before metric computation.

Static attributes: The 16 static attributes (one vector per basin) were normalized using training-basin statistics and tiled along the time axis so that each timestep of the input sequence carries both dynamic and static information. This simple concatenation approach avoids a separate static encoder while still injecting catchment characteristics. But it does mean more data redundancy and a larger input dimension for the sequence model to process.

Sequence windows: Overlapping 365-day sliding windows were extracted from each basin’s time series within each split. This produces approximately $50 \times 8,400 \approx 420,000$ training samples, $50 \times 1,100 \approx 55,000$ validation samples, and $50 \times 1,450 \approx 72,500$ test samples.

2 Problem 2: Model Design

Two sequence model architectures were implemented and compared.

2.1 LSTM

The LSTM model is a two-layer stacked LSTM followed by a linear output head:

1. **Input:** $\mathbf{X} \in \mathbb{R}^{B \times 365 \times 21}$
2. **Layer 1:** LSTM (input = 21, hidden = 128, dropout = 0.2)
3. **Layer 2:** LSTM (input = 128, hidden = 128, dropout = 0.2)
4. **Head:** Dropout(0.2) $\rightarrow$ Linear(128 $\rightarrow$ 1)
5. **Read-out:** Last hidden state $\mathbf{h}_T$ at $t = 365$.

Total trainable parameters: **208,513**.

Static attributes enter the model by being concatenated to each timestep of the dynamic input sequence, so the LSTM processes 21 features per step rather than 5.

Appropriateness for streamflow: LSTMs were first applied to rainfall-runoff by Kratzert et al. (2018) and have since become a benchmark. The gated recurrent structure acts as a learned conceptual store, naturally simulating the role of soil moisture, groundwater, and snowpack.

Strength: Handles long-range dependencies and captures seasonality through selective forgetting; well-studied for hydrology.

Weakness: Sequential computation cannot be parallelized over the time dimension; a fixed-size hidden state may compress information as the sequence length grows.

2.2 Transformer

The Transformer model uses a linear embedding layer, sinusoidal positional encoding, and a stack of two pre-norm TransformerEncoder layers, followed by mean-pooling and a linear output head:

1. **Input projection:** Linear(21 $\rightarrow$ 128)
2. **Positional encoding:** Sinusoidal (Vaswani et al., 2017)

3. **Encoder $\times 2$:** TransformerEncoderLayer ($d_{\text{model}} = 128$, $n_{\text{head}} = 4$, $d_{\text{ff}} = 256$, dropout = 0.2, pre-norm)
4. **Pooling:** Mean over the time dimension
5. **Head:** Dropout(0.2) $\rightarrow$ Linear(128 $\rightarrow$ 1)

Total trainable parameters: **267,905**.

Static attributes are injected the same way as for the LSTM.

Appropriateness: The self-attention mechanism can in principle attend to arbitrary lag relationships without distance decay, which could be advantageous for capturing remote snowmelt or monsoon precursors.

Strength: Fully parallel training; attention can explicitly link precipitation events to delayed flow responses anywhere in the 365-day context.

Weakness: Attention complexity is $O(T^2)$ in sequence length; with only 50 basins the Transformer may lack the data needed to learn meaningful attention patterns compared to the LSTM’s stronger inductive bias toward sequential dynamics.

3 Problem 3: Training Pipeline

3.1 Training Configuration

Both models were trained with the same hyperparameter configuration, summarized in Table 2. The CLI exposes all of these as command-line arguments (e.g., `--lr`, `--epochs`, `--hidden-size`).

Table 2: Training hyperparameters shared by both models.

Hyperparameter	Value
Sequence length	365 days
Batch size	256
Optimizer	Adam
Learning rate	1×10^{-3}
Loss function	MSE (normalized space)
Max epochs	30
Early stopping	patience = 10 epochs (best val loss)
Dropout	0.2
Hidden size / d_{model}	128
Number of layers	2
Random seed	42

The LSTM converged in **23 epochs** (best validation NSE = 0.889); the Transformer ran the full **32 epochs** before early stopping (best validation NSE = 0.836).

3.2 Training Curves

Figure 3 shows training and validation loss curves and validation NSE for both models. The LSTM converges smoothly and reaches a lower validation loss, with validation NSE climbing steadily to

≈ 0.89 . The Transformer converges more slowly with greater epoch-to-epoch variability in validation loss, reflecting the architecture’s weaker inductive bias for sequential data.

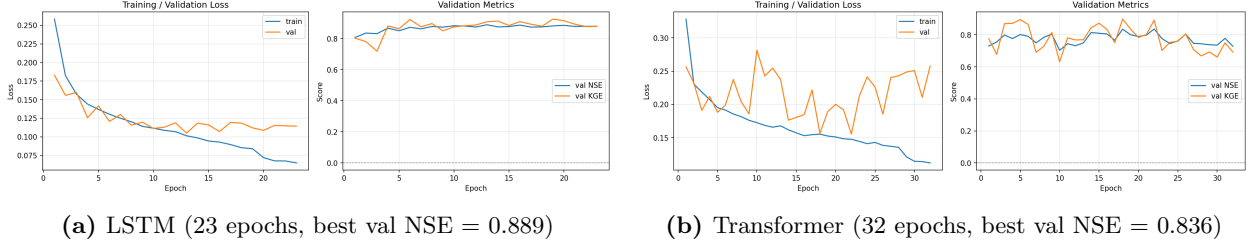


Figure 3: Training and validation loss and NSE curves for both models. The LSTM converges faster and achieves a higher peak validation NSE.

4 Problem 4: Evaluation and Interpretation

4.1 Test-Set Metrics

Both models were evaluated on the held-out test set (WY2007–WY2010) across all 50 basins. Table 3 reports the median of each metric. The LSTM substantially outperforms the Transformer on every metric.

Table 3: Test-set performance (median across 50 basins). PBIAS is reported as the median of the absolute value.

Model	NSE $\uparrow$	KGE $\uparrow$	RMSE $\downarrow$	MAE $\downarrow$	—PBIAS— $\downarrow$
LSTM	0.778	0.774	0.909	0.420	13.1%
Transformer	0.560	0.603	1.508	0.689	18.2%

The LSTM achieves $\text{NSE} > 0.5$ (“satisfactory” by Moriasi et al. 2007 guidelines) for **84%** of basins and $\text{NSE} > 0.7$ for **62%**. The Transformer achieves $\text{NSE} > 0.5$ for only **58%** of basins and $\text{NSE} > 0.7$ for just **20%**, underscoring the large performance gap.

Figure 4 visualizes the full empirical CDFs of all five metrics. The LSTM CDF is consistently shifted toward better values across every metric.

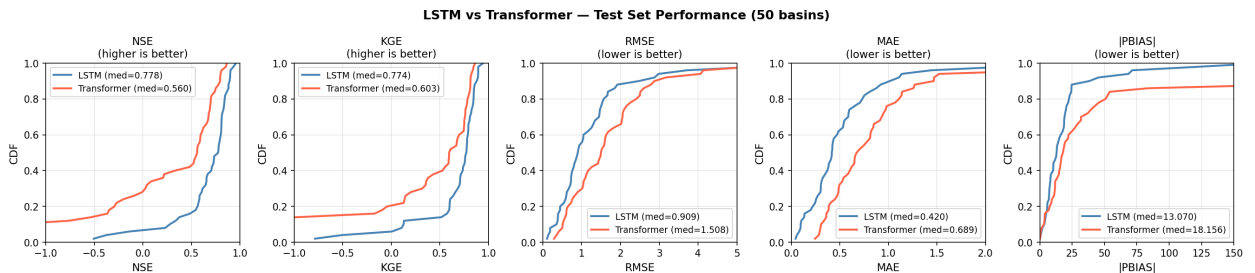


Figure 4: Empirical CDFs of test-set performance metrics for the LSTM (blue) and Transformer (red) across all 50 basins. Median values are shown in the legend. PBIAS is the absolute value; x-axes are clipped to highlight the central distribution.

4.2 Best-Performing Basin (LSTM): 12358500

Basin 12358500 achieved the highest LSTM test NSE of **0.960** ($KGE = 0.895$, $PBIAS = +8.6\%$). This basin is located in the Pacific Northwest (northern Rockies / Cascades region), a snowmelt-dominated catchment with a pronounced and predictable spring runoff peak. The strongly seasonal, single-peaked hydrograph provides clear temporal structure that the LSTM’s gated recurrence can exploit: the forget gate learns to retain snowpack accumulation signals over winter while the input gate activates during the spring melt period. Figure 5 shows close agreement between simulated and observed hydrographs throughout the test period, with the model accurately capturing both the timing and magnitude of peak flows.

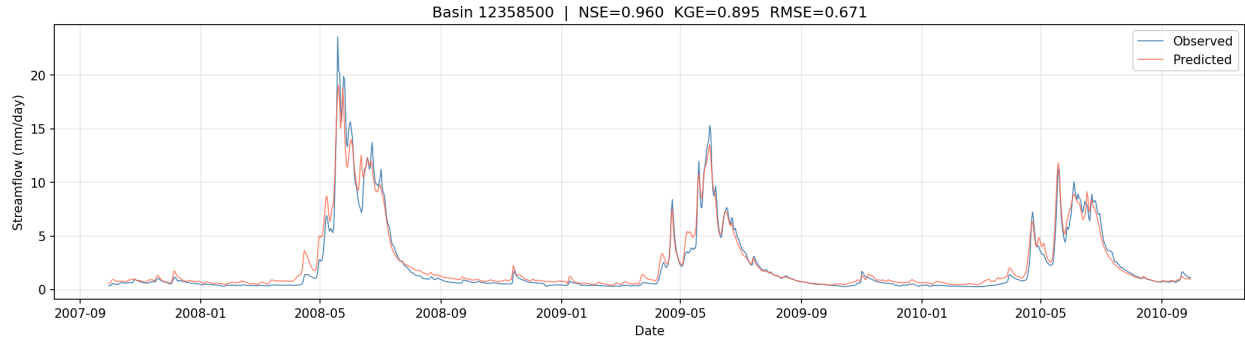


Figure 5: LSTM hydrograph for best basin (12358500, $NSE = 0.960$). Observed (blue) and simulated (orange) daily streamflow over the WY2007–WY2010 test period.

4.3 Worst-Performing Basin (LSTM): 02297310

Basin 02297310 had the worst LSTM test NSE of **−0.507** ($KGE \approx 0.006$, $PBIAS = +69\%$). This basin is located in the low-gradient coastal plain of Florida and is characterized by groundwater-dominated, flashy streamflow with a weak or absent seasonal signal. The positive bias indicates systematic overestimation: the model likely generalizes the precipitation-driven runoff dynamics it learned from energy-limited basins in the training set, while failing to account for the highly permeable soils and rapid subsurface drainage typical of this karst-influenced region. Lacking explicit soil or geology information beyond what is encoded in the 16 static attributes, the model cannot distinguish this basin’s behavior from a more typical rainfall-runoff regime. Figure 6 illustrates the poor correspondence, with simulated peaks greatly exceeding the relatively muted observed response.

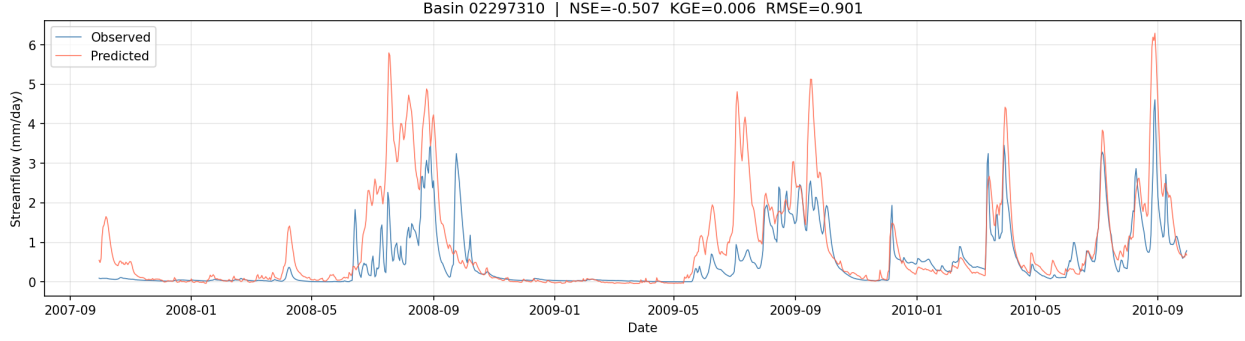


Figure 6: LSTM hydrograph for worst basin (02297310, $\text{NSE} = -0.507$). The model systematically overestimates streamflow, suggesting it fails to represent the unique groundwater-dominated regime of this coastal-plain basin.

4.4 Why the LSTM Outperforms the Transformer

The LSTM’s advantage can be attributed to three factors:

1. **Inductive bias:** The recurrent gated structure of an LSTM is a natural fit for the sequential memory processes (soil moisture routing, snowpack accumulation/melt) that govern rainfall-runoff. This prior structure reduces the amount of data needed to learn a useful representation. The Transformer, by contrast, must learn the relevance of temporal ordering entirely from scratch via positional encodings and attention patterns.
2. **Data scale:** With only 50 basins and 23 training years, the dataset is relatively small for a Transformer, which typically requires large data volumes to train its attention heads to produce physically meaningful lag structures. Several basins with atypical hydrology (e.g., semi-arid intermittent streams) appear to confuse the Transformer badly, as evidenced by catastrophic NSE values (basin 07148400: $\text{NSE} = -6.3$, $\text{PBIAS} = -417\%$) that suggest near-random predictions.
3. **Optimization stability:** The noisier validation loss curves of the Transformer (Figure 3) indicate a harder optimization landscape with the given learning rate and batch size. Transformer models often benefit from warmup schedules and larger batch sizes that were not explored here.

5 Conclusion

This assignment implemented a reproducible CLI-based rainfall-runoff prediction system using the miniCAMELS dataset. Two sequence architectures were trained and evaluated on 50 basins:

- The **LSTM** produced strong results (median test $\text{NSE} = 0.778$, $\text{KGE} = 0.774$), consistent with its well-established role as a benchmark model in data-driven hydrology. The model performed particularly well in snowmelt-dominated basins with clear seasonal structure and struggled most with flat, groundwater-dominated coastal-plain basins where its learned runoff relationships did not transfer.
- The **Transformer** underperformed substantially (median test $\text{NSE} = 0.560$), with several basins showing catastrophic failure. This is likely attributable to limited training data and the lack of the recurrent inductive bias that makes LSTMs well-suited to hydrological time series.

Future work could explore larger datasets, learning rate warmup, or hybrid architectures that combine attention with recurrence.

The key lesson is that architectural inductive biases matter significantly at data scales typical in catchment hydrology: the LSTM's built-in assumption of temporal ordering and memory provides a meaningful advantage over the more general Transformer when training data are limited.

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